

From Independent Scientist to Novel Performer

A Small-Business Operating, Milestone, and Capitalization Plan for a Phase 1 LLM-Advised Robotic Whipple (ChemicalQDevice)

Draft 1.0

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20 figures

12 milestones

SBIR Phase I

\$306,000

9 months, 5 milestones

SBIR Phase II

\$1,300,000

24 months, 7 milestones

Five-year programme

\$3,500,000

60 months, direct

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Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference.

Disclaimer: This work is independent and is not endorsed or sponsored by any trial sponsor, CRO, site, IRB, regulator, or medical society; and was adapted using Claude Code Opus 5.

The paper (v1.0) is at <https://doi.org/10.5281/zenodo.xxxxxxxx> and the repository (v4.5.0) is at <https://github.com/kevinkawchak/physical-ai-oncology-trials>. The paper files are deposited in funding/capitalization-plan.

Keywords Small business innovation research; Novel performers; Capitalization plan; Milestone schedule; Indirect cost; 21 CFR part 54; Pancreatic ductal adenocarcinoma; Robotic pancreaticoduodenectomy; Daraxonrasib; Physical AI

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Abstract

In August 2026 the author filed ten funding applications for a Phase 1 first-in-human trial of an LLM-advised robotic pancreaticoduodenectomy with perioperative daraxonrasib, written throughout as an independent scientist [?]. Nine of the ten address mechanisms that fund a person or an institution. One, application 05, addresses the only mechanism in the set built for a company, and it was the shortest of the ten. This paper converts the author from independent scientist to small-business operator and rewrites that application as the document it should have been.

The clause the conversion turns on is neither the report's individual-scientist language nor its novel-performers chapter. It is the Small Business Innovation Research clause of *Science: A New Golden Age*, which appears in three separate chapters and is the only clause in the report written for a firm [?]. Three numbers govern what follows: \$306,000 for a nine-month Phase I, \$1,300,000 for a twenty-four-month Phase II, and the \$3,500,000 five-year direct programme the two are measured against [?]. Two more are derived: \$1,396,000 of direct work sits inside the \$1,606,000 award, and \$2,104,000 sits outside it.

What is offered is a costed twelve-milestone schedule. Every row carries an evidence artifact that exists as a file, a cost that sums into its phase, a stop condition stated before the work begins, and the funder mechanism that pays.

Executive Summary

The clause. *Science: A New Golden Age* names SBIR three times: in the Chapter I summary of recommendations, in Chapter III on deploying SBIR and STTR strategically to couple federally-seeded companies with the scientific enterprise, and in Chapter IV on SBIR opening doors for technician-founded ventures. Its novel-performers chapter describes work at tens of millions of dollars with teams of ten to a hundred people over half a decade. This company proposes \$1,606,000, 2.6 full-time equivalents, and 33 months.

The entity. ChemicalQDevice holds thirteen deposited assets outright and none under encumbrance: two protocols, an IND package, a simulation stack, a verification suite, and a public repository. It licenses three things, none exclusively. It has contracted for nothing. Seven instruments are absent, and not one can be obtained by the company alone.

The two prices. The identical four-layer budget costs \$3,500,000 of direct work over sixty months and \$1,396,000 of direct work over thirty-three. The award that carries the second is \$1,606,000, a 15.0 percent overhead and fee load against an SBIR allowance of 40 percent that the company does not claim. Routed through a university at a 57 percent facilities and administrative rate, the same direct work would cost a funder \$2,137,000.

The bridge. Federal money of \$1,606,000 sits below a firewall; contributed drug, theatre and pathology sit unpriced in the middle; and \$5,900,000 of private capital sits above, raised only after a milestone closes. That is 3.67 to one on cash alone, against the annexed FY 2028 target of at least three to one [?].

The milestones. Twelve rows, five in Phase I summing to \$306,000 and seven in Phase II summing to \$1,300,000. Each is closed by three concurrent evidence branches that must join before an audit can compare them.

What the conversion costs. The individual-scientist framing that made all ten prior applications distinctive is abandoned. Capitalization detail the author might prefer private is published permanently under a DOI. Three of the ten prior recipients will read a company plan as off-mechanism. Those four concessions are stated in §9 and are not softened anywhere.

How to Read This Document

The document is written so that four different readers can each open it at a different page and find the thing they came for within two pages.

Reader	Open at	Then	The question this reader is actually asking
SBIR program director	§3, Table 8	§5, Tables 13 and 14	Does the award buy a defensible gate, and does the budget sum?
ARPA-H programme manager	§5, Figure 13	§1, Figure 1	Is there a milestone I can hold someone to, and a stop condition?
Private investor	§4, Table 12	§2, Table 6	What do I own, when may I own it, and what is already built?
UC San Diego intake	§8, Figure 19	§7, Figure 18	What is being asked of this institution, and by when?

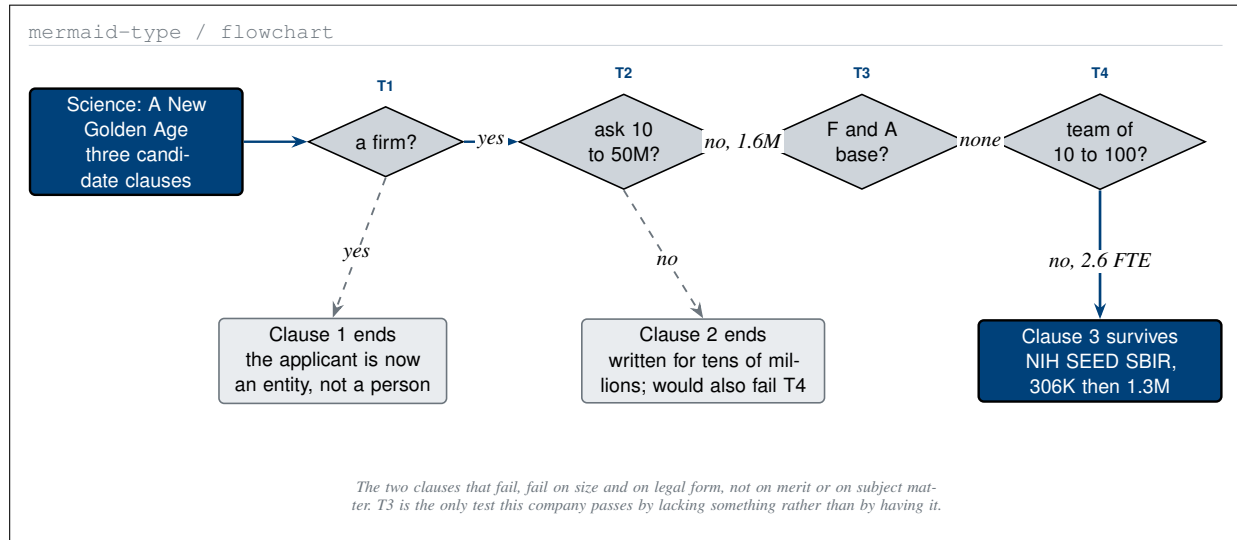
Four readers and the section each should open first. No reader is asked to read the document front to back, and every entry point is a table or a figure rather than a paragraph.

Fig	§	Platform	Source directory	The question it answers
1	1	Mermaid	mermaid/	Which clause does a company with no indirect-cost base qualify under?
2	1	D2	d2/	What kind of institution is a 2.6 FTE firm, on the report's own axes?
3	1	Graphviz	graphviz/	What does the same direct work cost under three overhead regimes?
4	2	Diagrams	diagrams-pyhton/	What is the shape of what the company holds today?
5	2	D2	d2/	What is owned, licensed, contracted, and absent, with terms?
6	2	PlantUML	plantuml/	What may the sponsor do alone, and what needs an institution?
7	3	Mermaid	mermaid/	What must hold at month nine for Phase II to begin?
8	3	D2	d2/	What does the same programme cost under two mechanisms?
9	3	Graphviz	graphviz/	Which work packages does the money reach, and which not?
10	4	PlantUML	plantuml/	Which capital positions may be held while a participant is on study?
11	4	D2	d2/	Where does every dollar come from, and what separates the tiers?
12	4	Mermaid	mermaid/	In what order is a private round executed during enrollment?
13	5	Mermaid	mermaid/	When does each milestone open, close, and produce its artifact?
14	5	Graphviz	graphviz/	What has to fail, and in what combination, to stop the programme?
15	5	PlantUML	plantuml/	What do three parties do at once to close one milestone?
16	6	D2	d2/	Which published quantities can a funder check before committing?
17	6	Graphviz	graphviz/	Where does each published number land in a filing?
18	7	Diagrams	diagrams-pyhton/	Who is employed, who is contracted, who is contributed?
19	8	Mermaid	mermaid/	What did the July and August 2026 contacts actually unlock?
20	10	Diagrams	diagrams-pyhton/	What survives if the programme stops, and who holds it?

The twenty figures, their section, their platform, the directory that carries each specification, and the one question each answers. Five mermaid, three plantuml, five d2, three diagrams, four graphviz, by purpose not quota.

Tab	§	Contents	The question it answers
1	0	Four readers, four entry points	Which section should this reader open first?
2	0	Figure index, twenty rows	Where is each figure and what does it answer?
3	0	Table index, nineteen rows	Where is each table and what does it answer?
4	1	Three clauses against four eligibility tests	Which clause survives the filter, and why?
5	1	Three overhead regimes on one direct base	What is the premium for routing through an institution?
6	2	The owned register, thirteen rows with DOIs	What exists today and where is it deposited?
7	2	The absent register, seven rows with signatories	What is missing and who alone can supply it?
8	3	The four-layer budget under two prices	What does the SBIR route reach, and what does it not?
9	3	Phase I line items, eight rows to \$306,000	What does nine months of Phase I actually buy?
10	3	Phase II line items, ten rows to \$1,300,000	What does twenty-four months of Phase II buy?
11	4	The §54.2 and §54.4 disclosure triggers	What must be disclosed, at what threshold, on which form?
12	4	The capitalization table across five stages	Who owns what, and does the SBIR test still pass?
13	5	The twelve milestones, costed and dated	What is the milestone, when, and what does it cost?
14	5	Artifact, stop condition, funder mechanism	What proves it, what halts it, and who pays?
15	6	Six checkable quantities with limitations	What can a reviewer verify against a published source?
16	6	Cost and schedule against industry benchmarks	What did the prior work cost, and what does it cost elsewhere?
17	7	The 2.6 FTE by employment status	Who is on the payroll and who is not?
18	9	Risks, likelihood, and the mitigation held	What could go wrong and what is already in place?
19	11	Abbreviations, two column pairs	What does this term mean in this paper?

The nineteen tables, their section, their contents, and the one question each answers. Every table is set to the full body measure with ragged-right fixed columns, so no cell shows a stretched interword gap.



Three clauses of one report, four eligibility tests, and the only clause a San Diego firm with no indirect-cost base survives. The two that fail, fail on size, not on merit or on subject matter.

1 The Novel-Performer Case

1.1 Three Clauses, One Applicant

Science: A New Golden Age contains three clauses under which a Phase 1 pancreatic cancer trial could plausibly be funded, and they are not interchangeable [?].

The first is the individual-scientist language of Chapter I, which asks the roughly \$200 billion annual federal research and development portfolio to stop deferring to “the same incumbents that consume the funding”. The ten prior applications were written under it [?].

The second is the chapter section headed NOVEL PERFORMERS, which describes “a vast middle ground of mid-scale science”, problems “requiring tens of millions of dollars, coordinated teams of ten to a hundred people, and timelines of half a decade”.

The third is the Small Business Innovation Research clause, and it is the only one of the three that appears in more than one chapter. The Chapter I summary of recommendations asks agencies to “focus Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs to build strategic capabilities”. Chapter III states that “SBIR and STTR programs can be deployed strategically to advance new scientific and technological capabilities, coupling federally-seeded companies with the scientific enterprise”. Chapter IV states that “Programs like SBIR open doors for technician-founded ventures, directing resources toward ideas that do not always start with a dissertation”.

Test	Question	Basis in the report	ChemicalQDevice, August 2026
T1	Is the applicant a firm rather than a person?	Chapter IV, SBIR opens doors for technician-founded ventures	Yes, a California corporation
T2	Is the ask ten to fifty million?	Chapter II, novel performers is written for tens of millions	No, \$1,606,000 over 33 months
T3	Is there an indirect-cost base to recover?	Chapter II, deference to the incumbents that consume the funding	No negotiated rate agreement exists
T4	Is the team ten to a hundred people?	Chapter II, coordinated teams of ten to a hundred	No, 2.6 FTE at full Phase II staffing

The four tests, the sentence in the report that each is drawn from, and this company's answer. Clause 1 fails T1, clause 2 fails T2 and T4, and clause 3 is the only one that survives all four.

1.2 Why the Pioneer Language Does Not Fit an Entity

What changed between the ten applications and this one is the applicant. A person-based mechanism funds a named scientist and follows that person's judgement for five to seven years. ChemicalQDevice is a California corporation with one officer, and a corporation is not a subject that mechanism has a way to fund. Nothing about the science changed; the legal form of the applicant did.

That is a real loss and it is worth naming here rather than in a footnote. The individual-scientist framing is what made all ten prior applications distinctive, and it is the framing the report itself rewards most explicitly. Abandoning it is the first of four costs the conversion carries, and all four are set out in §9 without mitigation.

1.3 Why the Novel-Performers Chapter Does Not Fit Either

The novel-performers chapter is precise about scale, and the precision is what excludes this company. It describes problems requiring tens of millions of dollars, teams of ten to a hundred people, and timelines of half a decade, and it names focused research organizations and the NSF TIP X-Labs as the vehicles built for them. Set against that: \$1,606,000, 2.6 full-time equivalents, and 33 months.

The chapter describes a real gap in the federal portfolio. This company does not fill it, and claiming otherwise would be exactly the kind of overreach the report is written against.

Two well suited, three partly suited, two less suited. The two well-suited rows are public-goods data and infrastructure, and mission-driven public-good science, and both are well suited for the same structural reason: every one of the twelve milestone artifacts in §5 is deposited publicly by plan, and the programme has one mission with a stop condition and a closure budget.

A firm claiming to be well suited to all seven would be claiming to be a university, a corporate laboratory, a federal laboratory and a focused research organization at once. The report's own table exists to make that claim impossible to state.

d2-type / true grid

Activity	University	Corporate lab	Federal lab	New institutions	SBIR firm 2.6 FTE
Curiosity driven, investigator led	Well suited	Less suited	Partly suited	By design	Partly suited
Larger scale, engineering intensive	Less suited	Partly suited	Partly suited	By design	Less suited
Long horizon platform and tools	Less suited	Partly suited	Well suited	By design	Partly suited
Public goods data and infrastructure	Partly suited	Less suited	Partly suited	By design	Well suited
Mission driven, public good science	Partly suited	Less suited	Partly suited	By design	Well suited
Proprietary product development	Less suited	Well suited	Less suited	By design	Partly suited
Workforce training and apprenticeship	Well suited	Partly suited	Less suited	By design	Less suited

☐ Well or partly suited, the report's own scoring

☐ Less suited

☒ Well suited, this company

The rule at the right marks where the report's own columns end and the column this paper adds begins. The four left-hand columns are the report's scoring restated in words, because its plus, tilde and cross glyphs are not legible at this measure.

The report's own seven activities, its four institutional forms, and a fifth column it does not carry. An SBIR firm of 2.6 people is well suited to two of the seven and unsuited to two others.

graphviz-type / record nodes

Direct work, identical		Route A, university		Route B, full allowance		Route C, this plan	
Personnel, 2.6 FTE	\$842,000	Direct	\$1,396,000	Direct	\$1,396,000	Direct	\$1,396,000
Site clinical conduct	\$288,000	Less equipment	-\$96,000	Indirect, 40 pct	\$558,000	Indirect, 7.5 pct	\$105,000
Compute and rig	\$96,000	MTDC base	\$1,300,000	Subtotal	\$1,954,000	Subtotal	\$1,501,000
Monitoring and DSMB	\$86,000	F and A, 57 pct	\$741,000	Fee, 7 pct	\$137,000	Fee, 7 pct	\$105,000
Consultants, travel	\$84,000	Fee	none				
Total direct	\$1,396,000	Total to funder	\$2,137,000	Total to funder	premium \$485,000, 30.2 percent		\$1,606,000



Every column begins at the same \$1,396,000 and every total is the sum of its own column. The company may claim route B and does not; the difference between B and C is \$485,000 that is never requested.

The same 1,396,000 of direct work priced three ways. A university F and A rate at 57 percent adds 741,000, the full SBIR allowance adds 695,000, and the rate this plan claims adds 210,000 in all.

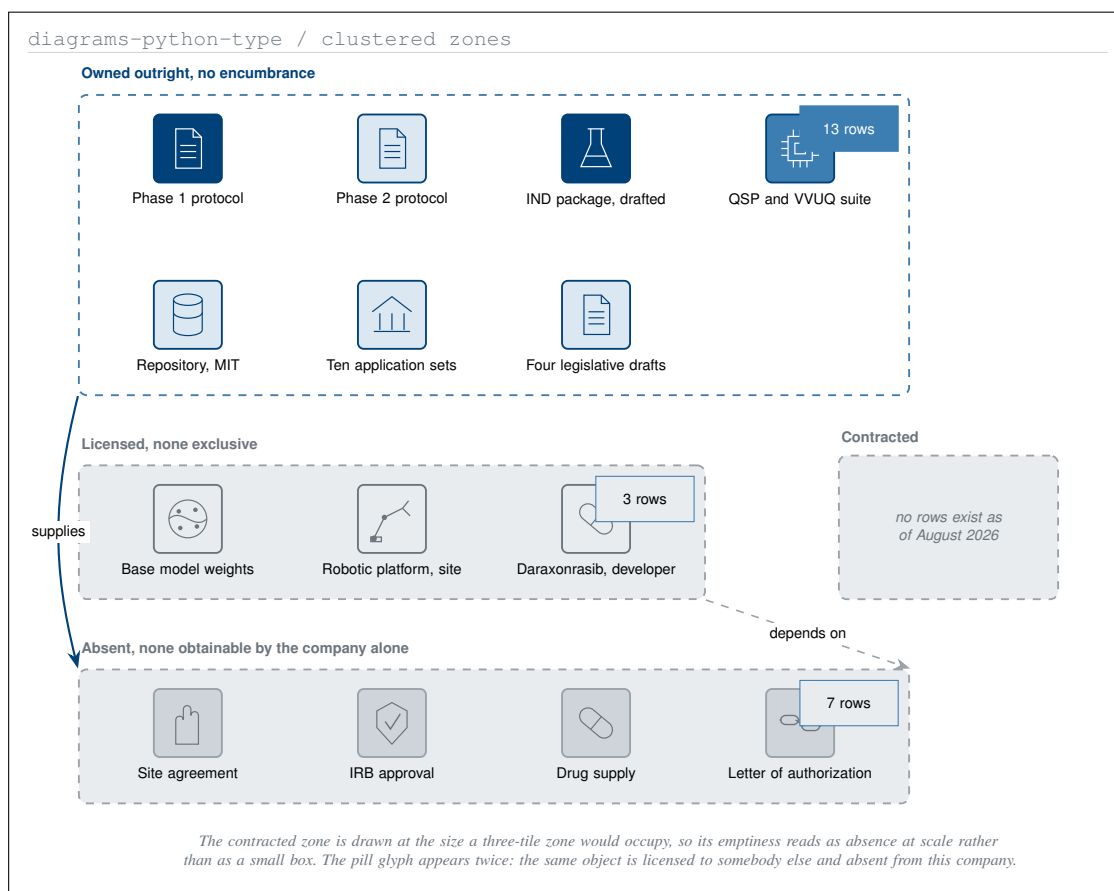
1.4 The Clause That Fits, and What It Is Worth

ChemicalQDevice has no negotiated indirect-cost rate agreement, no facilities to recover, and no administrative layer to fund. That is a structural fact about a 2.6 person firm rather than a concession, and it has a price a program officer with a fixed appropriation can read directly.

Line	Route A, university	Route B, full allowance	Route C, this plan
Direct costs	\$1,396,000	\$1,396,000	\$1,396,000
Equipment removed from base	-\$96,000	not applicable	not applicable
Overhead base	\$1,300,000 MTDC	\$1,396,000 direct	\$1,396,000 direct
Overhead rate	57 percent	40 percent	7.5 percent
Overhead	\$741,000	\$558,000	\$105,000
Fee at 7 percent	none	\$137,000	\$105,000
Total to the funder	\$2,137,000	\$2,091,000	\$1,606,000
Premium over route C	\$531,000, 33.1 pct	\$485,000, 30.2 pct	zero

The same direct work under three overhead regimes. The 57 percent rate is a typical negotiated university rate; the 40 percent is the SBIR allowance available without a rate agreement; the 7.5 percent is what this firm documents.

The 7.5 percent rate covers insurance, audit and accounting, and nothing else, because there is nothing else to cover [?]. The company could claim the 40 percent allowance without documentation and does not [?]. For a program officer holding a fixed appropriation the consequence is arithmetic rather than rhetorical: the same trial, run by the same person, costs \$531,000 less than it would through an institution and \$485,000 less than this mechanism permits.



Four zones and thirteen tiles. The owned zone holds everything the company made; the contracted zone is drawn at full size and holds nothing, because an empty zone is the finding this figure reports.

2 The Entity and the Asset

2.1 What the Entity Is

ChemicalQDevice is a California corporation. Its sole officer, sole director and sole owner is Kevin Kawchak. It has no employees at the date of writing, no negotiated indirect-cost rate agreement, and no facilities of its own. It is therefore more than 50 percent directly owned and controlled by one United States individual, which is the eligibility test the SBIR programme applies [?], and §4 is written to keep it so through the thirty-three months this plan covers.

2.2 Owned

Asset	Instrument	First deposit	Identifier
Phase 1 protocol	Zenodo	2026-06-21	10.5281/zenodo.20780121
Phase 2 protocol	Zenodo	2026-06-24	10.5281/zenodo.20807027
IND package, drafted, not filed	Zenodo	2026-07-01	10.5281/zenodo.21097442
Phase 1 LLM document guidance	Zenodo	2026-06-29	10.5281/zenodo.21018646
PI LLM adoption guide	Zenodo	2026-06-26	10.5281/zenodo.20843290
QSP simulation stack and VVUQ suite	Zenodo	2025-08	10.5281/zenodo.17001137
Daraxonrasib identification meta-analysis	Zenodo	2025-06	10.5281/zenodo.15735068
Funding application v1.0	Zenodo	2026-07-07	10.5281/zenodo.21232965
Funding application v2.0	Zenodo	2026-07-12	10.5281/zenodo.21317266
Patient robot advocacy paper	Zenodo	2026-07-31	10.5281/zenodo.21720120
H. R. 9510 bill v5.0 and companions	Zenodo	2026-06-12	10.5281/zenodo.20619762
Repository, MIT licensed	GitHub	2026-08	physical-ai-oncology-trials v4.5.0
Ten funding application file sets	Repository	2026-08-04	funding/pdac-funding-applications

Thirteen owned assets, every one deposited and addressable. No row carries an encumbrance, no row is subject to a licence granted to a third party, and no row was produced under a grant that reserves rights in it.

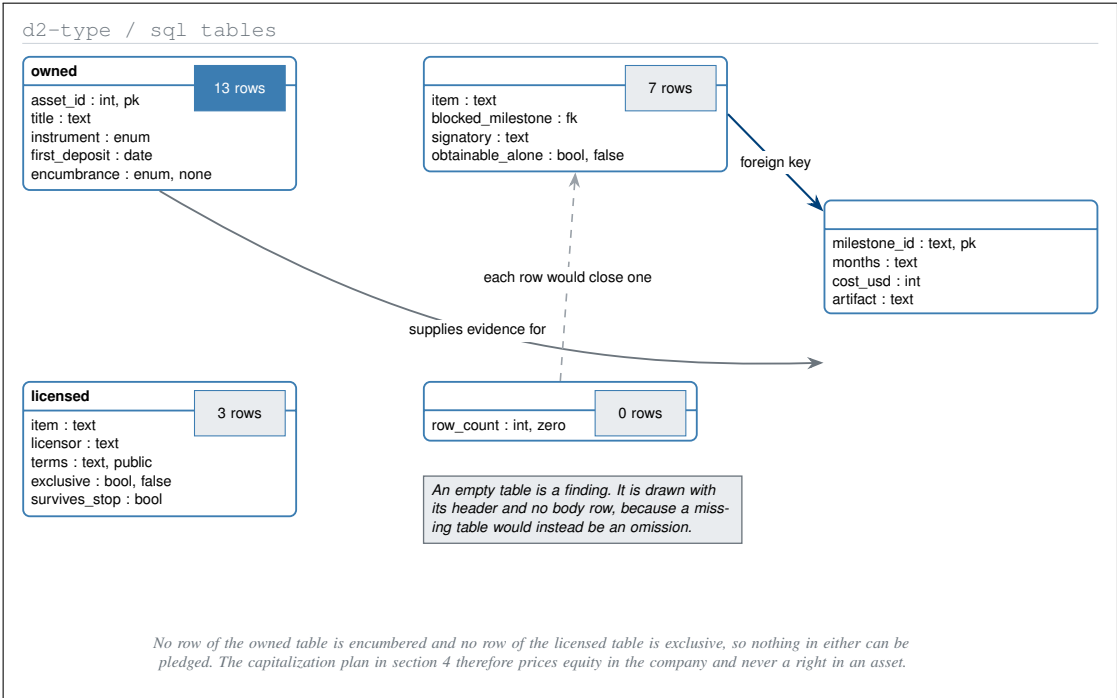
What the thirteen rows cost to produce is on the record and is unusual enough to state. The empirical triplicate, the ten-arm quantitative systems pharmacology virtual trial, and the digital-twin platform were each produced for an estimated \$36,330 and in about one month, against industry benchmarks of above \$120,000, above \$2,000,000, and \$28,000 to \$700,000 respectively [?]. Those figures are estimates from the source set and are reported as such, not as audited accounts; Table 16 carries them with their benchmarks.

The consequence for a funder is that none of the thirteen appears in the budget. The protocol is written, the simulation stack is built, the verification suite is frozen, and the IND package is drafted [?, ?, ?, ?]. A request for \$1,606,000 is a request to run a trial, not to write one.

2.3 Licensed, and Contracted

Three things are licensed and none exclusively. The base model weights are used under their vendor's public terms. The robotic platform is the site's asset and not the company's. Daraxonrasib is the developer's investigational agent, and no licence of any kind exists. None of the three survives a stop, because none is owned.

The contracted table has no rows. No clinical trial agreement, no drug supply agreement, no letter of authorization, no confidentiality agreement, no subaward, and no trial-specific insurance binder is in force.



The asset register as four typed records. Thirteen rows are owned outright, three are licensed on public terms, seven are absent, and the contracted table has no rows at all as of August 2026.

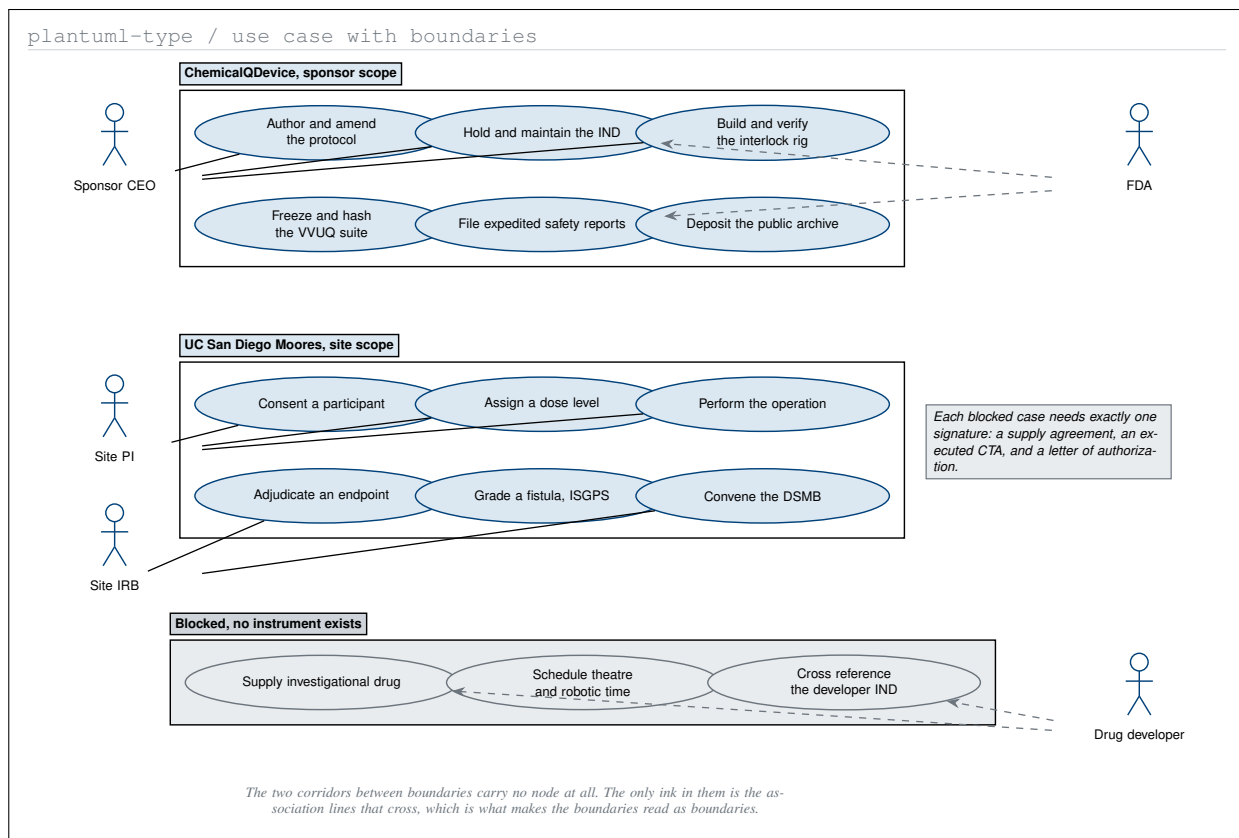
2.4 Absent

Instrument absent	Blocks	Only signatory	Obtainable by the company alone
Site agreement, executed	M1	UC San Diego	No
IRB approval	M2	Site institutional review board	No
Drug supply agreement	M6	Revolution Medicines	No
Letter of authorization, IND cross-reference	M5	Revolution Medicines	No
Theatre and robotic platform time	M7	UC San Diego	No
Trial-specific liability insurance	M6	Carrier	No, needs an executed CTA
A second full-time employee	M3	Funded by the Phase I award	No, needs the award

Seven absent instruments, the milestone each blocks, and the single party that can supply it. The last column reads no on every row, which is the whole content of the table and the reason §8 exists.

Three use cases sit in the corridor, and each is blocked by exactly one signature. No amount of company work removes any of the three, which is the honest reading of the register and the reason the twelve milestones in §5 open with a site feasibility memorandum rather than with an experiment.

The register is dated. On 13 January 2026, when the first proposal was released, the owned table held two rows and the absent table held the same seven [?]. In the seven months since, the owned table has grown by eleven rows and the absent table has not lost one. That asymmetry is the single most useful thing in this section: it says precisely which kind of work the author can do alone and which kind he cannot.



Two system boundaries, twelve use cases, and the corridor between them holding three that neither party may execute today. Each of the three is blocked by exactly one missing signed instrument.

3 Placeholder

4 Placeholder

5 Placeholder

6 Placeholder

7 Placeholder

8 Placeholder

9 Placeholder

10 Placeholder

11 Placeholder
